

## SIXTH SEMESTER U.G. DEGREE EXAMINATION, MARCH 2022

(CBCSS-UG)

Mathematics

MTS 6B 14 (E02)—TOPOLOGY OF METRIC SPACES

(2019 Admissions)

Time : Two Hours

Maximum : 60 Marks

## Section A

*Answer at least eight questions.**Each question carries 3 marks.**All questions can be attended.**Overall Ceiling 24.*

1. Define metric space.
2. Define Isometry between metric spaces.
3. Define Norm on a Linear Space over  $\mathbb{R}$  or  $\mathbb{C}$ .
4. Find all limit points of  $S = \left\{ \frac{1}{n} : n \in \mathbb{N} \right\}$  in  $\mathbb{R}$  with Euclidian metric.
5. Suppose  $X$  is a metric space and  $A \subseteq B \subseteq X$ . Prove that  $\text{acc}(A) \subseteq \text{acc}(B)$ .
6. Prove that any function from a Discrete metric space to a metric space is continuous.
7. Suppose  $X, Y$  and  $Z$  are metric spaces and  $f : X \rightarrow Y$  and  $g : Y \rightarrow Z$ . If  $f$  and  $g : Y \rightarrow Z$  are continuous, prove that  $g \circ f : X \rightarrow Z$  is continuous.
8. Suppose  $X$  and  $Y$  are metric spaces and  $f : X \rightarrow Y$ . Show that  $f$  is continuous at every isolated point of  $X$ .
9. Define open map. Give one example.
10. Define Connected metric space. Give one example.
11. Find all isolated points of  $S = \left\{ \frac{1}{n} : n \in \mathbb{N} \right\}$  in  $\mathbb{R}$  with Euclidian metric.
12. Define Cauchy sequence. Give one example.

(8 × 3 = 24 marks)

Turn over

### Section B

*Answer at least five questions.*

*Each question carries 5 marks.*

*All questions can be attended.*

*Overall Ceiling 25.*

13. Prove that continuous image of a connected metric space is connected.
14. Prove that Every convergent sequence in a metric space is Cauchy. Give one example of a sequence which is not Cauchy
15. Suppose  $(X, d)$  be a metric space and  $\hat{d}(x, y) = d(x, y)/(1 + d(x, y))$  for each  $y \in X$ . Show that  $\hat{d}$  is a metric on  $X$ .
16. Suppose  $X$  is a metric space and  $A$  and  $B$  are subsets of  $X$  for which  $A \subseteq B$ . Prove that  $\text{diam}(A) \leq \text{diam}(B)$ .
17. Suppose  $X$  is a metric space and  $S$  is a subset of  $X$ . Prove that  $\partial S = \partial(S^c)$ .
18. Prove that  $\mathbb{Q}$ , the set of rationals, is dense in  $\mathbb{R}$ . Give one example of infinite subset of  $\mathbb{R}$  which is not dense in  $\mathbb{R}$ .
19. Suppose  $X$  is a non-empty metric space,  $z \in X$  and  $S \subseteq X$ . Prove that following statements are equivalent :
  - (i)  $\text{diam}(S) < \infty$ .
  - (ii) There is a ball of  $X$  centred at  $z$  that includes  $S$ .
  - (iii) There is a ball of  $X$  that includes  $S$ .

(5 × 5 = 25 marks)

### Section C

*Answer any one question.*

*The question carries 11 marks.*

20. Suppose  $X$  and  $Y$  are metric spaces  $f : X \rightarrow Y$  and Prove the following statements are equivalent :
  - (i) For each open subset  $V$  of  $Y$ , the inverse image  $f^{-1}(V)$  is open in  $X$ .
  - (ii) For each closed subset  $F$  of  $Y$ , the inverse image  $f^{-1}(F)$  is closed .
  - (iii) For each open ball  $B$  of  $Y$ , the inverse image  $f^{-1}(B)$  is open in  $X$ .
21. Suppose  $S$  is a subset of  $\mathbb{R}$ . Prove that  $\text{diam}(S) = \sup S - \inf S$ .

(1 × 11 = 11 marks)